The 2025-26 San Ramon sequence: SED's rtDD and DL picker with the 2015 settings

2026-10-18

Summary

We repeated the 2015 San Ramon test on the 2025-26 San Ramon sequence (California), with unchanged settings:

- SED's double-difference relocator rtDD, both as scrtdd (SeisComP) and as pyrtdd (Python);
- SED's deep-learning picker scdlrtpicker.

The data are public NCEDC data: 927 events, 41,870 analyst and automatic picks, and waveforms of 26 test events.

- rtDD runs unchanged and gives the same answer in both forms.
  - scrtdd and pyrtdd agree to a median of 1 m, and 97.8 % of their cross-correlation lags are identical.
  - 893 of the 927 events were relocated. The 34 dropped events have fewer than 8 picks, and most of them are unreviewed.
- rtDD sharpens the routine catalog about 2.4 times, but cross-correlation helps much less than in 2015.
  - Against Lamont's real-time double-difference catalog, the local geometry is off by a median of 142 m horizontal and 217 m vertical (2015: 51 / 63 m against Lamont's cross-correlation re-analysis).
  - The causes cannot be separated with these data:
    - the yardstick is weaker this time (a real-time solution, with events twice as sparse);
    - the sequence is wider;
    - the station network is different;
    - the waveform correlations are poorer.
- The new sequence did not reactivate the 2015 fault strand.
  - It lies on a parallel strand about 2 km east of the 2015 fault plane, at the same depth.
  - Its main body is 1.8-4.7 km SSE of the 2015 swarm along strike.
  - Only 8 of its 893 relocated events lie within 500 m of the 2015 plane.
- The DL picker behaves exactly as in 2015, and the data are now better for it.
  - On isolated events the PhaseNet models find the same share of analyst P picks in both years (0.89 for `phasenet_instance`).
  - Because the strong-motion records are now continuous, `phasenet_instance` finds 85 % of the analyst P picks at 3-component stations (2015: 56 %), with a median timing difference of about 0.03-0.05 s.
- What to watch for.
  - rtDD's absolute depth is again about 2.4 km too deep, so use only the relative geometry.
  - Unreviewed (automatic) events cause the few runaway relocations.
  - The picker loses one of two events that are only a few seconds apart.

1. The sequence and the data

- Timing. The swarm began in November 2025, peaked in November-December 2025 and February 2026, and continues at a low rate. Its largest event is the Mw 4.18 of 2026-02-02 15:01:13 UTC (nc75305736), 4 km SE of San Ramon.
- USGS forecast. USGS's aftershock forecast (issued 2026-10-12) treats it as a swarm ("a forecast based solely on aftershock processes can not capture all of the characteristics of the more complex

behavior of a swarm"). Its fitted model has p = 1.15 and b = 1.0, and it gave a 4 % chance of an M 3+ event in the following month.

	2015 swarm	2025-26 sequence
period	Oct-Nov 2015	2025-10-01 to 2026-10-18
events (NCSS routine or NCEDC DD location in the box)	685	927
picks used by rtDD (P / S)	25,684 (22,545 / 3,139)	41,870 (37,929 / 3,941)
events with automatic picks only	0	98
largest event	Mw 3.6 (nc72540765)	Mw 4.18 (nc75305736)
independent reference	Lamont NCAeqDD v202112.1 (cross-correlation re-analysis)	Lamont DD-RT v202201 (real-time DD)
velocity model	NCSS CON, Vp/Vs 1.73	the same (NCSS still uses CON here)

- Data used. Everything comes from NCEDC:
  - the NCSS catalog with phase picks, StationXML, and 62,246 cross-correlation windows;
  - for the picker, 26 test events with 4 min of data at all stations within 80 km.
- One notable difference from 2015: only 91 stations within 60 km are common to both years.

2. Relocation

- The runs. The same four runs as in 2015, with the same settings:
  - P0 / S0 used catalog times only;
  - P1 / S1 also used waveform cross-correlation (1.5-15 Hz, P on the vertical, S on both horizontals combined).
  - P = pyrtdd, S = scrtdd.
- What the distances mean.
  - ***Relative*** error: an event's position after removing each catalog's centroid.
  - ***Local*** error: the error in the vectors from each event to its 10 nearest neighbors, i.e. how well the small-scale structure is reproduced.

median difference from Lamont's DD catalog (m), horizontal / vertical	2015 relative	2015 local	2025-26 relative	2025-26 local
NCSS routine (the start)	229 / 394	292 / 506	307 / 730	365 / 853
rtDD, catalog times only (P0 = S0)	142 / 134	137 / 147	167 / 274	149 / 265
rtDD with cross-correlation (P1 = S1)	97 / 106	51 / 63	145 / 311 (233*)	142 / 217
NCEDC DD catalog, for scale	162 / 236	190 / 241	74 / 158	98 / 204

* with the median offset removed instead of the mean; about 16 runaway events shift P1's mean depth.

- rtDD again improves the routine catalog. 77 % of the events end closer to the reference than they started, 94 % in local geometry, and the 3-D improvement is about 2.4 times. But cross-correlation reduces the local errors by only 5 % (horizontal) and 18 % (vertical), where in 2015 it roughly halved them.
- Why we cannot say whether rtDD or the reference is at fault.
  1. A weaker yardstick. Lamont's real-time catalog places each new event against older reference events in real time; it is not a joint cross-correlation re-analysis like 2015's. Its events are also twice as far apart (210 m against 107 m).
  2. A wider sequence. Event pairs are a median of 1.04 km apart (2015: 0.77 km).

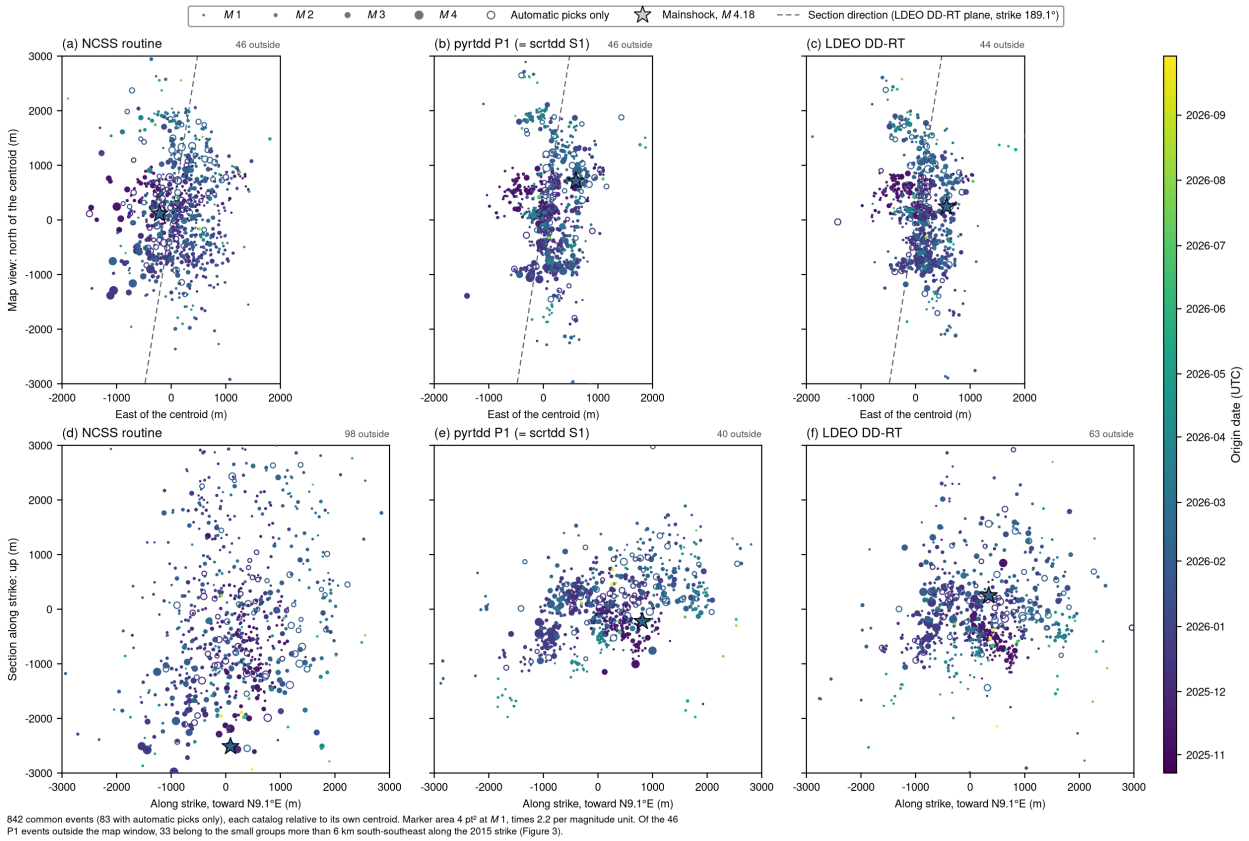


Figure 1: The sequence in three catalogs, each relative to its own centroid: the routine NCSS catalog, rtDD with cross-correlation (P1 = S1) and Lamont's real-time DD. rtDD shrinks the routine catalog's depth scatter to the level of the DD reference.

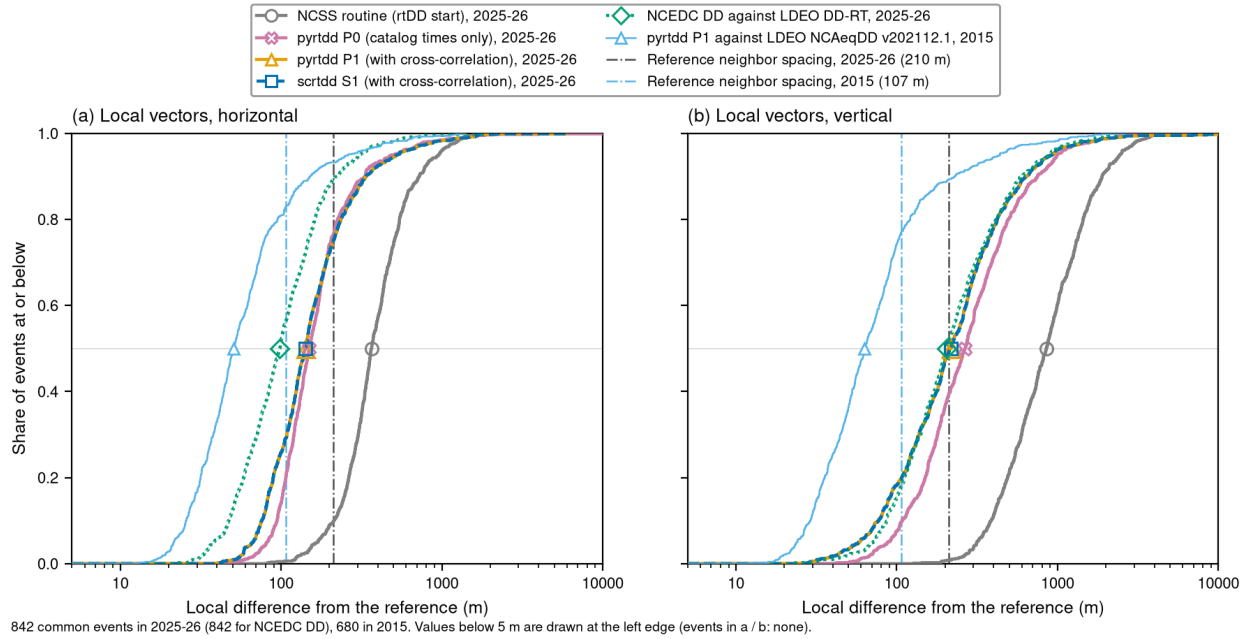


Figure 2: How close each catalog comes to Lamont's DD catalog in local geometry (cumulative distributions). The thin sky-blue curve is 2015's rtDD result against its own Lamont reference.

3. A different network, with fewer near-field stations.
4. Poorer correlations:
   - 54 % of the P pairs reach |CC| ≥ 0.7 (2015: 58 %), and 23 % of the accepted P values are negative;
   - the accepted lags are longer;
   - the final residual spread is 21 ms (2015: 13 ms).

- The two DD catalogs agree better with each other than with rtDD, but both are real-time solutions tied to the same base catalog, so that agreement is not an independent check.
- Absolute position. rtDD's default solver again puts the cluster about 2.4 km too deep and 1.3 km too far west. Only the relative geometry should be used.
- Runtime (one CPU core): about 1.5-2 min without cross-correlation, and 29 min (scrtdd) / 38 min (pyrtdd) with it.

3. Where the new sequence lies relative to the 2015 swarm

- Method. Each year's rtDD solution (P1) was shifted by its own mean offset against its Lamont reference, so both sit in Lamont's common frame.
- Check. The same picture follows from the two Lamont catalogs alone.

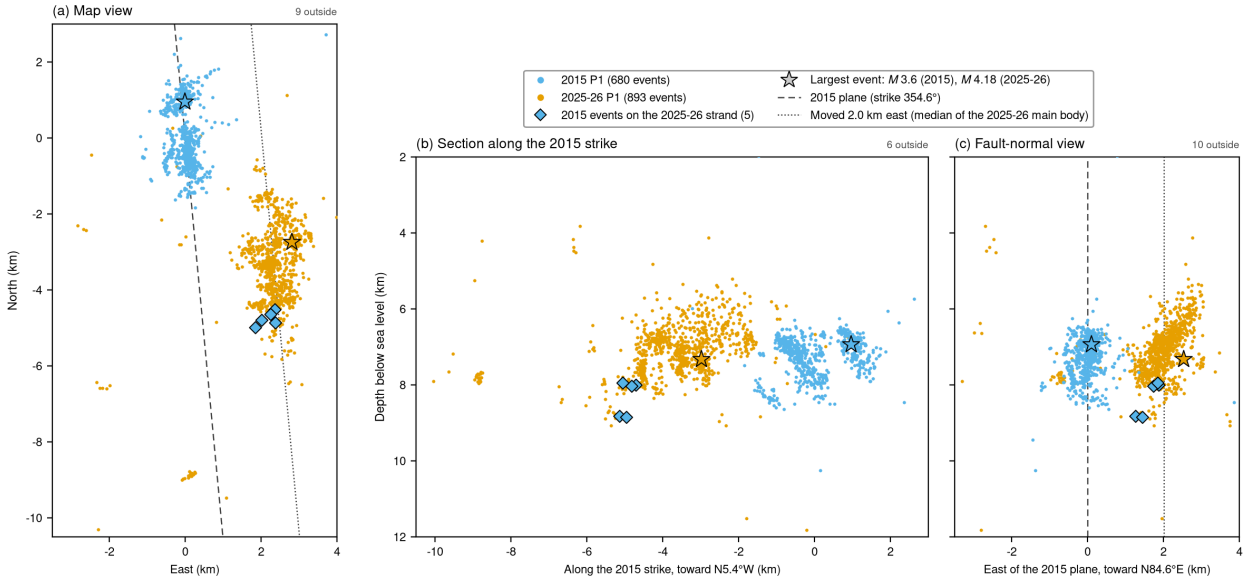


Figure 3: The 2015 swarm (blue) and the 2025-26 sequence (orange) in one frame: map view, a section along the 2015 fault strike, and a view across it. The 2025-26 sequence lies on a parallel strand about 2 km east of the 2015 plane; five 2015 events (diamonds) already lay on that strand.

- Position. The new centroid lies 4.1 km from the 2015 centroid, toward the SSE (azimuth 147°), at the same depth (7.1 km against 7.2 km).
- Not the 2015 strand. 846 of the 893 events lie more than 1 km east of the 2015 fault plane, and only 8 lie within 500 m of it.
- A parallel structure. The main body (835 events) lies on a structure parallel to the 2015 strand, within 2-5° of strike. A small cluster of five 2015 events at the south-east edge of that swarm already lay on this structure.
- Small groups further SW. 24 events sit about 8-9 km SW along strike at 7.8 km depth, and 9 shallower events sit at 4.5 km depth.

4. Deep-learning picker

- Setup. The same four models and settings as in 2015, on 26 analyst-reviewed events (M 0.6-4.18), compared with the analysts' picks.
- What recall means. Recall is the share of analyst picks the model also finds, within 0.5 s for P and 1.0 s for S.

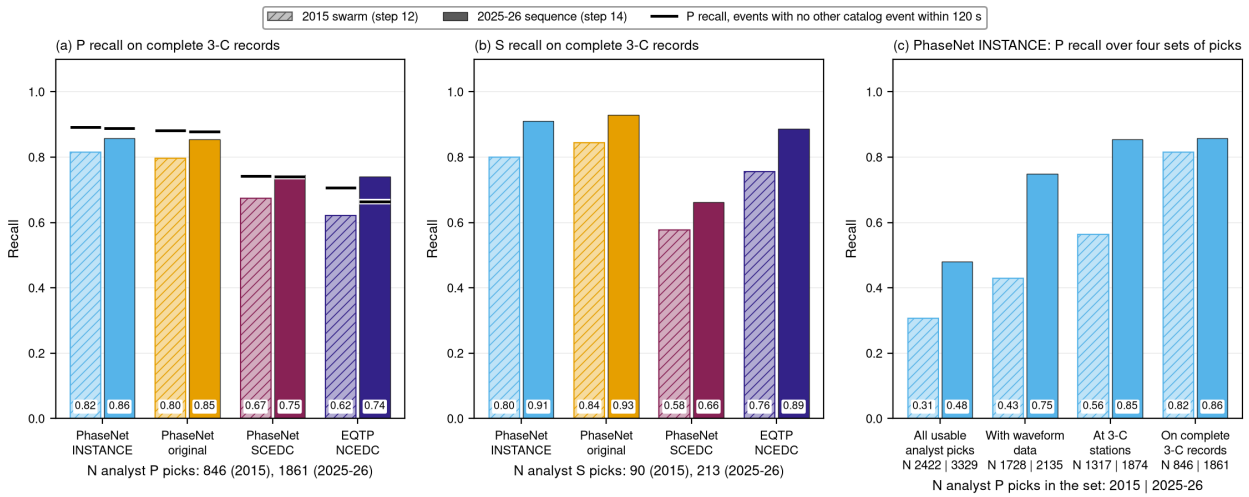


Figure 4: P and S recall on complete 3-component records, 2015 (hatched) vs 2025-26 (solid), with the recall on isolated events (black lines); (c) recall of phasenet_instance over four sets of analyst picks.

`phasenet_instance`	2015	2025-26
P recall: all analyst picks / with data / at 3-C stations / complete records	0.31 / 0.43 / 0.56 / 0.82	0.48 / 0.75 / 0.85 / 0.86
P recall, events with no other event within 120 s	0.89	0.89
S recall, complete records	0.80	0.91
false P picks per station-hour	1.08	0.87
complete station-event records	54 %	99 %

- Same models, changed data.
  - On isolated events the three PhaseNet models score exactly as in 2015 (0.89 / 0.88 / 0.74). The models behave the same.
  - In 2015, 46 % of the station records were triggered strong-motion records that started too late for the picker's detector; now 99 % of the records are continuous.
- Which model. `phasenet_instance` remains the best P picker, with the fewest false picks.
  - The fixed ranking rule names `phasenet_scedc` this time. It wins only because it makes the fewest S picks, scored against just 229 analyst S picks; it has the lowest P and S recall of the PhaseNet models.
- Close event pairs still lose one event. With one P per analysis window and a dead time, PhaseNet keeps the larger Mw 3.77 and loses the event 5.5 s before it.
- First-motion polarity (`eqtp_ncedc`, normally oriented channels).
  - At the plugin's default confidence of 0.5 it labels 88 % of the analyst first motions and agrees with 89 % of them.
  - At confidence 0.7 it labels 72 % and agrees with 93 %.
  - On boreholes with an inverted vertical channel, the analyst first motions follow the raw counts, so those 26 pairs disagree.
  - The model costs 22-25 times more CPU than PhaseNet.
- Data and network quirks found.
  - At the station nearest the swarm (NC.CSR), the binding rule picks a broadband sensor that started only on 2026-02-04. 20 earlier events lose that station, although its accelerometer picks well.
  - Three accelerometers have StationXML epochs that the picker cannot assemble into three components.

5. What this means

- The tools work unchanged on a new sequence, with a different station network, two years of archive and unreviewed events, and scrtdd and pyrtdd stay interchangeable. One caveat: stock pyrtdd still needs our filtering workaround.
- The cross-correlation gain is not guaranteed. It was large for the compact, well-recorded 2015 swarm against a re-analysis catalog. It is small here, and a stronger reference (a future Lamont re-analysis covering 2025-26) would be needed to judge rtDD's precision on this sequence.
- Practical settings for a sequence like this:
  - relocate only analyst-reviewed events, or at least check automatic ones;
  - judge rtDD only on relative geometry, or anchor it with better absolute travel times (station corrections);
  - use `phasenet_instance` for picking, and `eqtp_ncedc` only where polarity is needed, at confidence 0.7;
  - bind the sensor that covers the whole period at each station.
- Seismological result. The 2025-26 activity occurred on a separate, parallel strand of the San Ramon swarm zone, not on the 2015 one.

6. Caveats

- The reference. Lamont's real-time DD catalog is a real-time product, and NCEDC DD is tied to the same base catalog, so the two are not independent.
- Picks. 98 events have automatic picks only, and the near-station S picks arrive late against the model (median S-P residual about +0.3 s). These cancel in double differences but bias absolute depths.
- Picker sample. The picker test has 26 events (4 per magnitude bin), and the analyst picks are incomplete, so its precision is a lower bound.

7. Where to find everything

All paths are relative to the project folder (`/home/msseo/works/74.SeisComP_rtDD_DLpicker`).

- Clean notebooks (key figures only):
  - `notebooks/08_sanramon2026_relocation.ipynb`
  - `notebooks/09_sanramon2026_dlpicker.ipynb`
- Full result notes:
  - `docs/results/13_sanramon2026_relocation.md`
  - `docs/results/14_sanramon2026_dlpicker.md`
- Data preparation:
  - `docs/sanramon2026_inputs.md`
  - `docs/sanramon2026_waveforms.md`
  - the folder `data/sanramon_2026/`
- Run scripts and outputs: `runs/sanramon2026/`, with sub-folders `pyrtdd`, `scrtdd`, `analysis` and `dlpicker`.
- The 2015 test, for comparison: notebooks 06 and 07, and the notes `docs/results/11_*.md` and `12_*.md`.

Data: NCEDC (doi:10.7932/NCEDC); NCSS; Lamont-Doherty Earth Observatory double-difference catalogs; USGS ComCat.